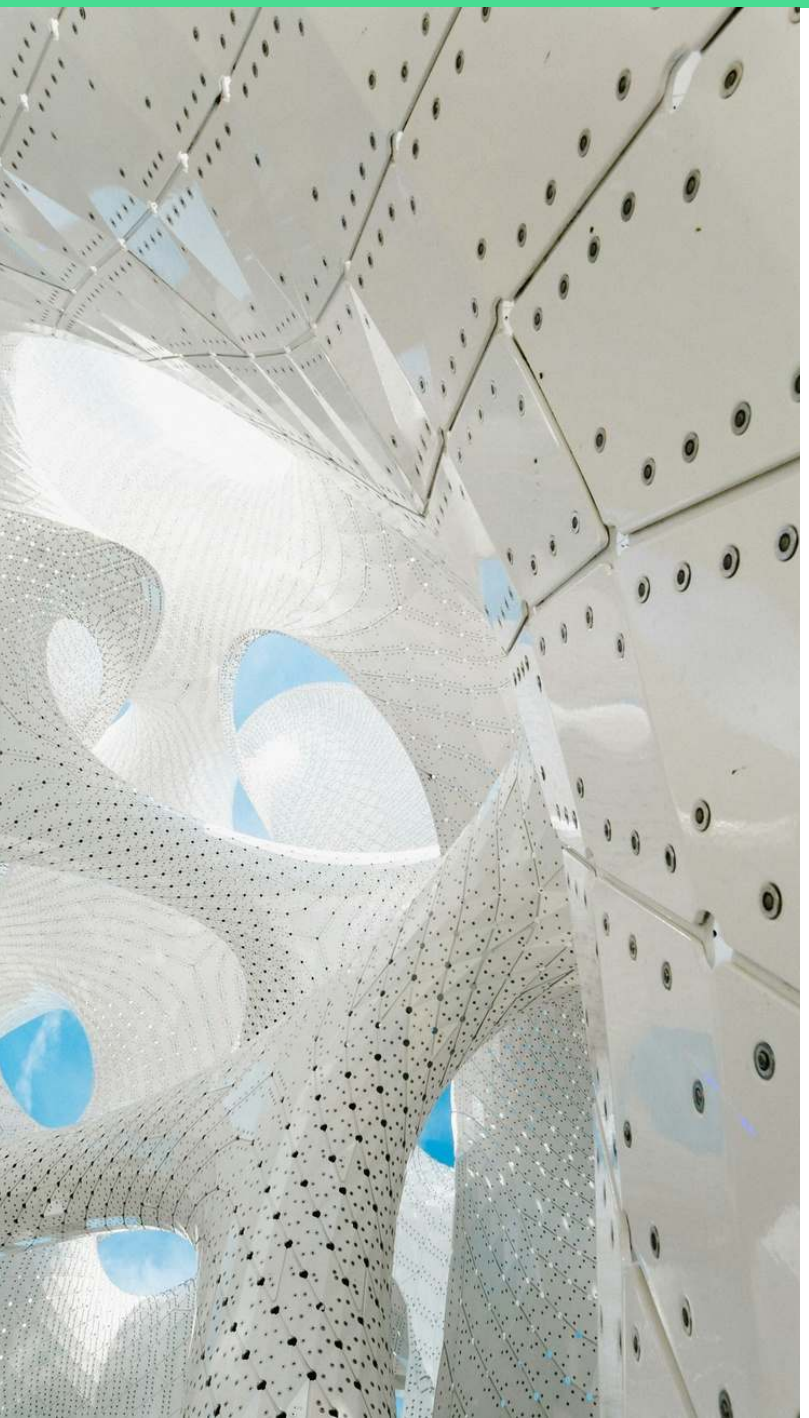


Extinction Archive: Umwelt Hypothesis Dossiers

AI memorial for lost species · sensory time capsule

Program Details

Team & Institution

- **Institution:** Macau University
- **Mentor:** Atticus SIMS
- GUO XIAO YUE — ID MC569254
- LIU JIA QUN — ID MC569293

Extinction Archive: Umwelt Hypothesis Dossiers

“Umwelt Hypothesis Dossiers”

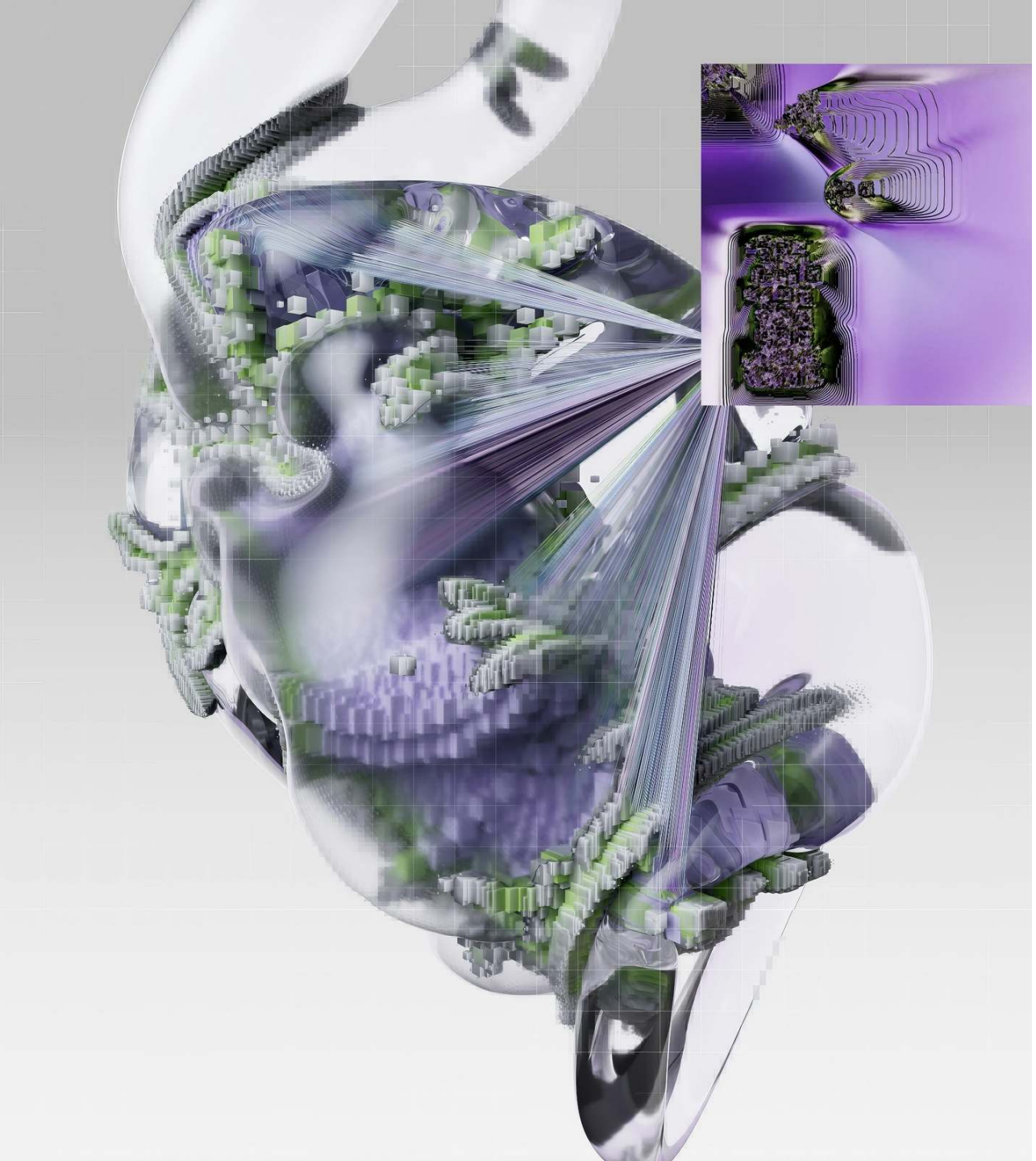
each species as a **folder of testable claims** (genes, orbits, climate), not a fantasy safari.

“AI memorial”

generative media **constrained** by literature — grief and wonder without faking science.

“Sensory time capsule”

we reconstruct **when** they lived (phase, season), not only **what** they looked like.



Premier student competition: biotechnology × art × design.

2026 gallery: “Convergent Life.” Targets: Biodigital Excellence; Narrative, Context, Reflection. Judging: biodesignchallenge.org/judging

Roles & Build Scope

1. **Experience lead**
   narrative, ethics, palaeo-chronobiology translation.

2. **Tech / AI lead**
   WebXR (A-Frame / Three.js), generative environments, epistemic UI.

3. **Strategy**
   重内容、轻装置 — Web-first; minimal tactile anchor + QR.

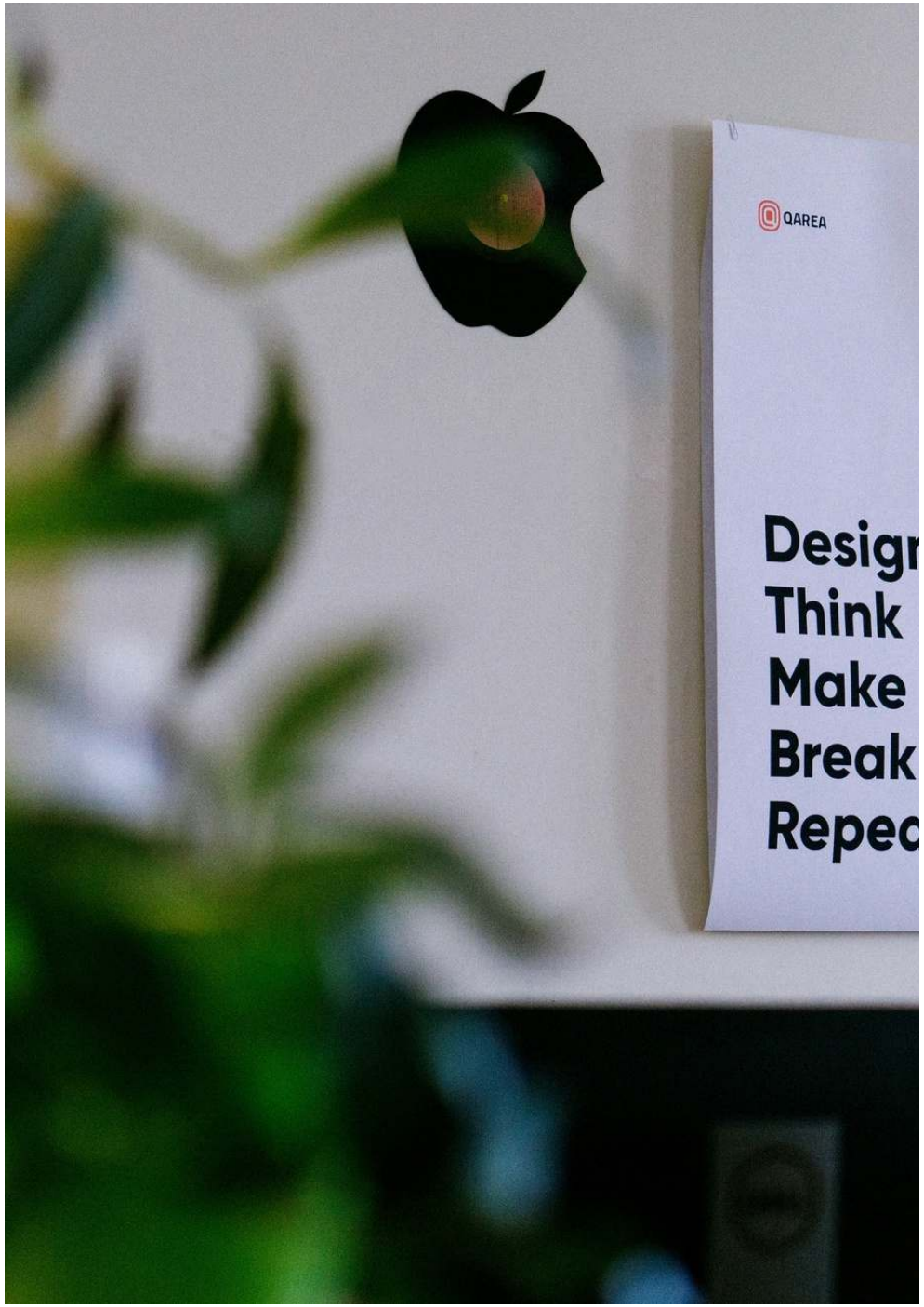
The Challenge of Extinction and Memory

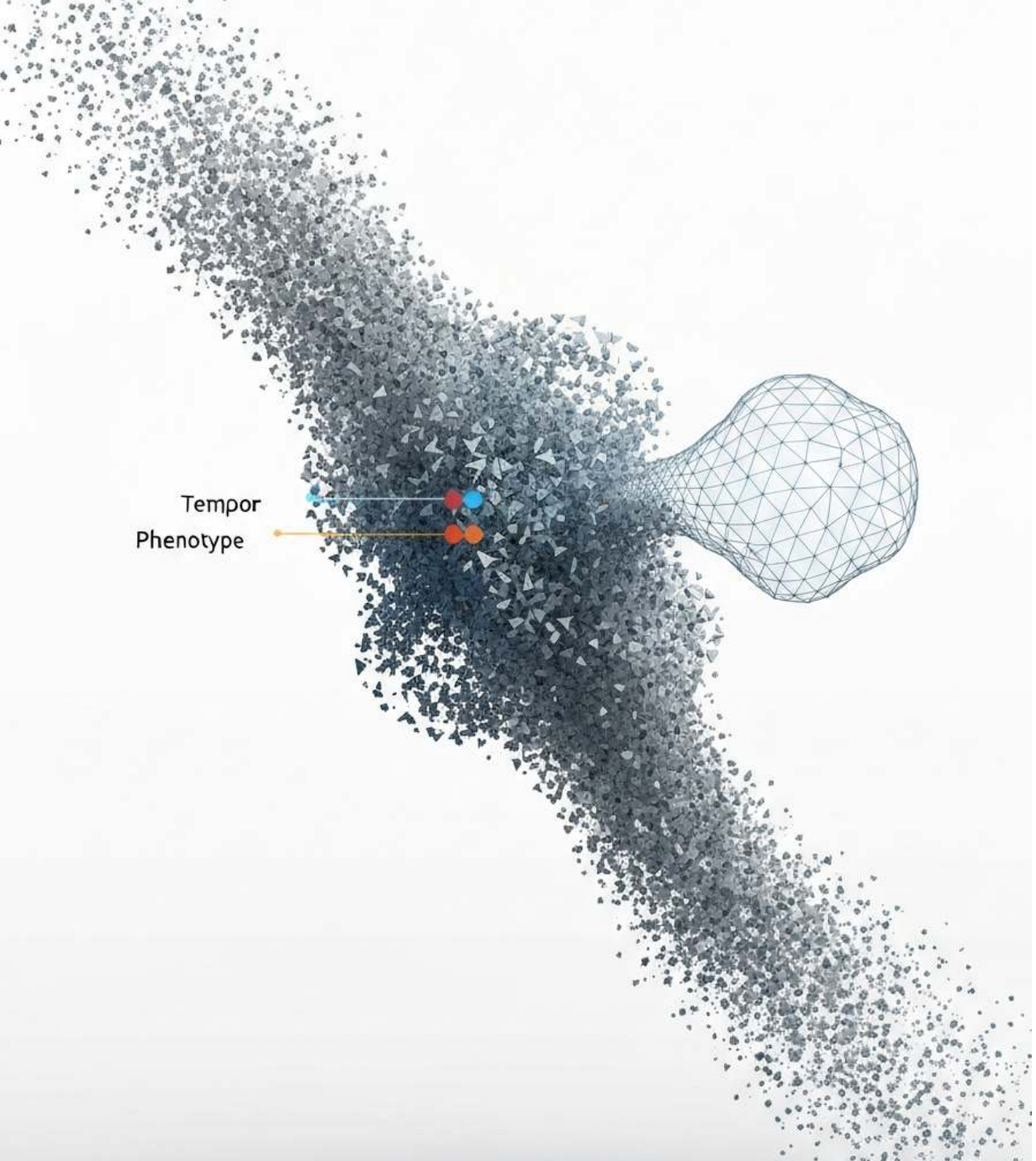
Problem → Provocation

Extinction erases **temporal niches** (daily / seasonal rhythms), not only DNA.

Public memory = flat images; we lack **felt time** of another species' world.

Provocation: can biodigital design surface that loss and force honest **de-extinction** debate?



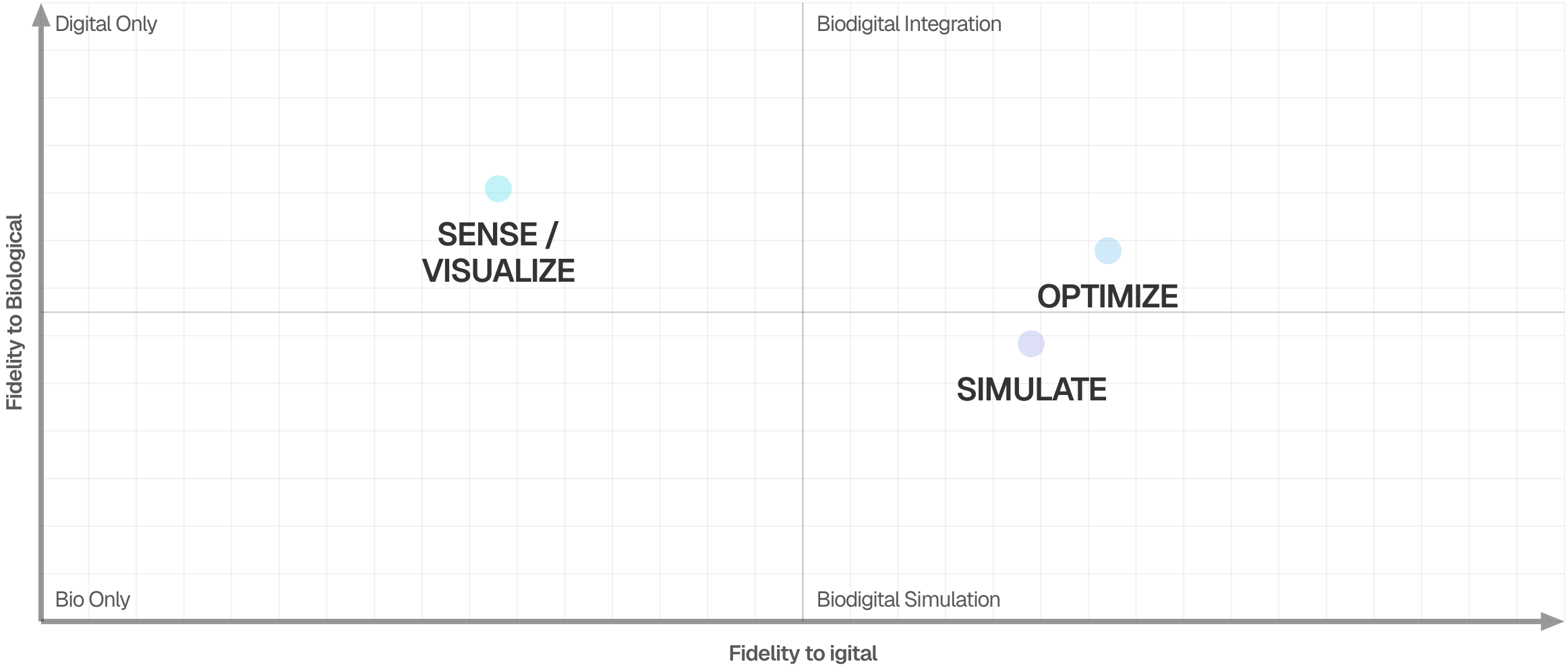


If we use generative AI + WebXR to reconstruct a temporal phenotype, can we repair the human gap in sensing extinction?

Palaeo-chronobiology + morphology (orbit, clock pathways, TRPV3...) = evidence ladder. Umwelt = sensory world as hypothesis, disclosed layer-by-layer.

Bio × Digital Quadrants

Our Mapping



Creative Concept — Two Dossiers Only

- **Woolly mammoth**

  Arctic circadian genomics (Lynch 2015), reindeer analogy (Lu 2010), mammoth steppe (Zimov 2012), TRPV3 cold (Lynch; Kim preprint optional).

- **Thylacine**

  skull / diel activity (Pozniak 2018), RGC metaphor (Mass & Supin 2020), history (Paddle; Sleightholme), sovereignty (Clements + Palawa-led Rimmer, Lehman; Schlunke).

- **Journey**

  discover rhythm → epistemic layers → ethics sliders → reflection log (HTML template in repo).

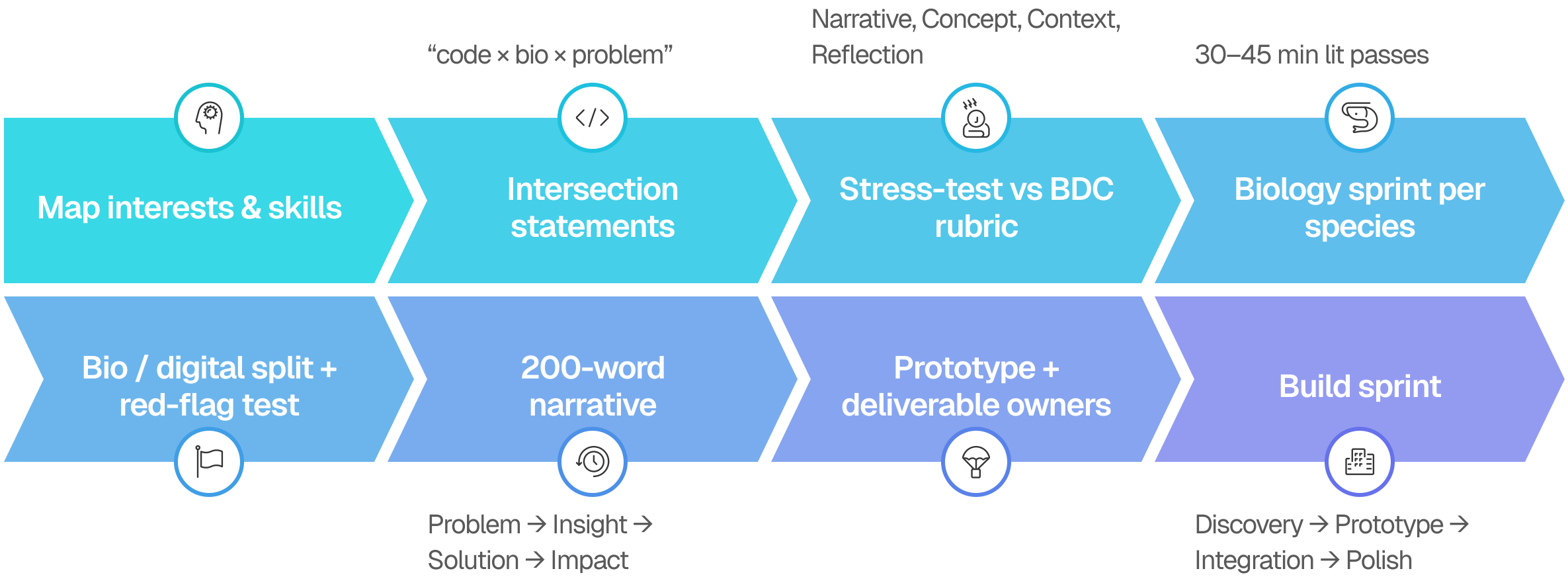
Scene IDs → Citations

- **Mammoth path**
- **Scene_Bio_01**
  Lynch (circadian-related genes)
- **Scene_Environment_02**
  Lu [Interpolated]
- **Scene_Landscape_01**
  Zimov
- **Scene_Sensory_01**
  TRPV3 / cold
- **Thylacine path**
- **Scene_Bio_02**
  Pozniak / orbit
- **Scene_POV_01**
  Mass & Supin [Interpolated]
- **Scene_Context_01**
  Paddle + Sleightholme
- **UI_Indigenous_Context**
- **UI_Ethics_Fork**
- **UI_Reflection_Log**

Design Techniques & Stack

- **WebXR**
  A-Frame / Three.js; browser-based Summit demo.
- **Generative AI**
  360° / environment art bounded by citation-backed parameters (no silent photoreal “truth”).
- **Epistemic UI**
  toggles for Cited / Interpolated / Speculative; on-screen (Author, year) tags.
- **Sonification (optional)**
  tie audio to phase / season from literature.
- **Reflection**
  `templates/reflection-log-webxr.html` — keyword → ethics cards (Sherkow & Greely themes).

Ideation Method (8 Steps)



Guided by `biodesign_cursor_agent.md` + course workbook.

Four-Week Production Spine

Week 1: Discovery

Literature grid, storyboard ↔ DOI table, WebXR shell.

Week 2: Prototype

Time scrubber, generative constraint doc, epistemic toggles.

Week 3: Integration

Sonification, Indigenous + ethics UI, reflection panel.

Week 4: Polish

10′ talk, 5′ Q&A drill, 1–5′ video, deploy, judge cheat sheet.

Context

Colonial Violence & Country

Thylacine loss tied to colonial harm to Palawa peoples and lutruwita (Tasmania).

Sources: Clements (2025); Schlunke (2025) poetry / Palawa names; **Palawa-led** Rimmer (2020), Lehman (2013).

UI protocol: author positionality; no tourism copy as TEK.

Ethics Layer

Sherkow & Greely (2013) - Grounded in Science

A **De-extinction vs. In Situ Conservation Funding**

Comparison of funding allocation between de-extinction efforts and established in situ conservation programs.

B **Legal Novelty and Risks**

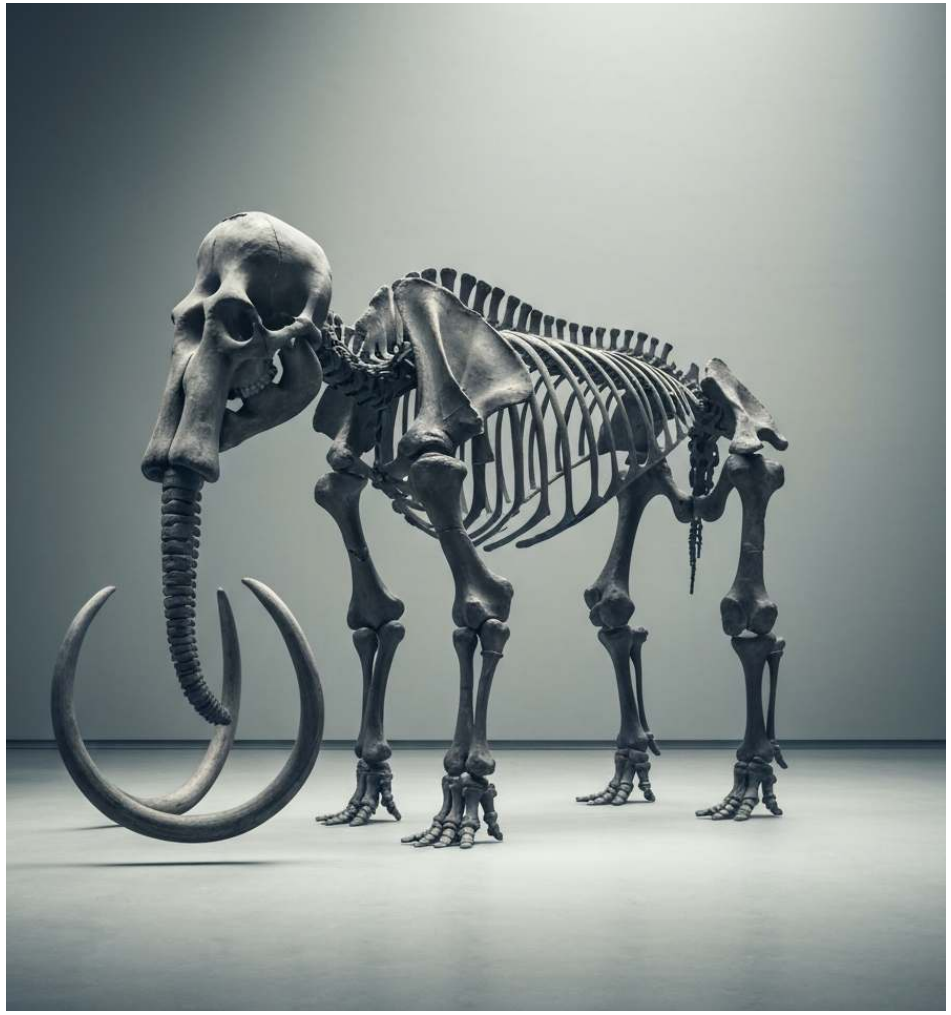
The legal implications and potential ecosystem risks associated with the introduction of revived lineages.

C **Published Research**

Research is grounded in scientific principles (DOI 10.1126/science.1236946), not speculative science fiction scenarios.

Grounded in Science DOI 10.1126/science.1236946 — not generic sci-fi branches.

Key Literature — Mammoth (DOIs)



- **M1 Lynch et al., 2015**
  10.1016/j.celrep.2015.06.027
- **M2 Lu et al., 2010**
  10.1016/j.cub.2010.01.042
- **M3 Zimov et al., 2012**
  10.1016/j.quascirev.2012.08.004
- **M4 Kim bioRxiv**
  10.1101/151746 (TRPV3 structure; phenotype cite Lynch 2015)
- **M5 Sherkow & Greely**
  10.1126/science.1236946

Key Literature — Thylacine & Sovereignty

- **Pozniak 2018**
  DOI: 10.1002/ar.23910
- **Mass & Supin 2020**
  DOI: 10.31857/S0002332920060107
- **Paddle 2000**
  ISBN: 9780521782196
- **Sleightholme & Campbell 2016**
  DOI: 10.1080/00222933.2016.1217037
- **Clements 2025**
  DOI: 10.1177/13534858251272203
- **Schlunke 2025**
  DOI: 10.1017/ext.2025.10008
- **Rimmer 2020 Artlink**
- **Lehman 2013 Griffith Review**

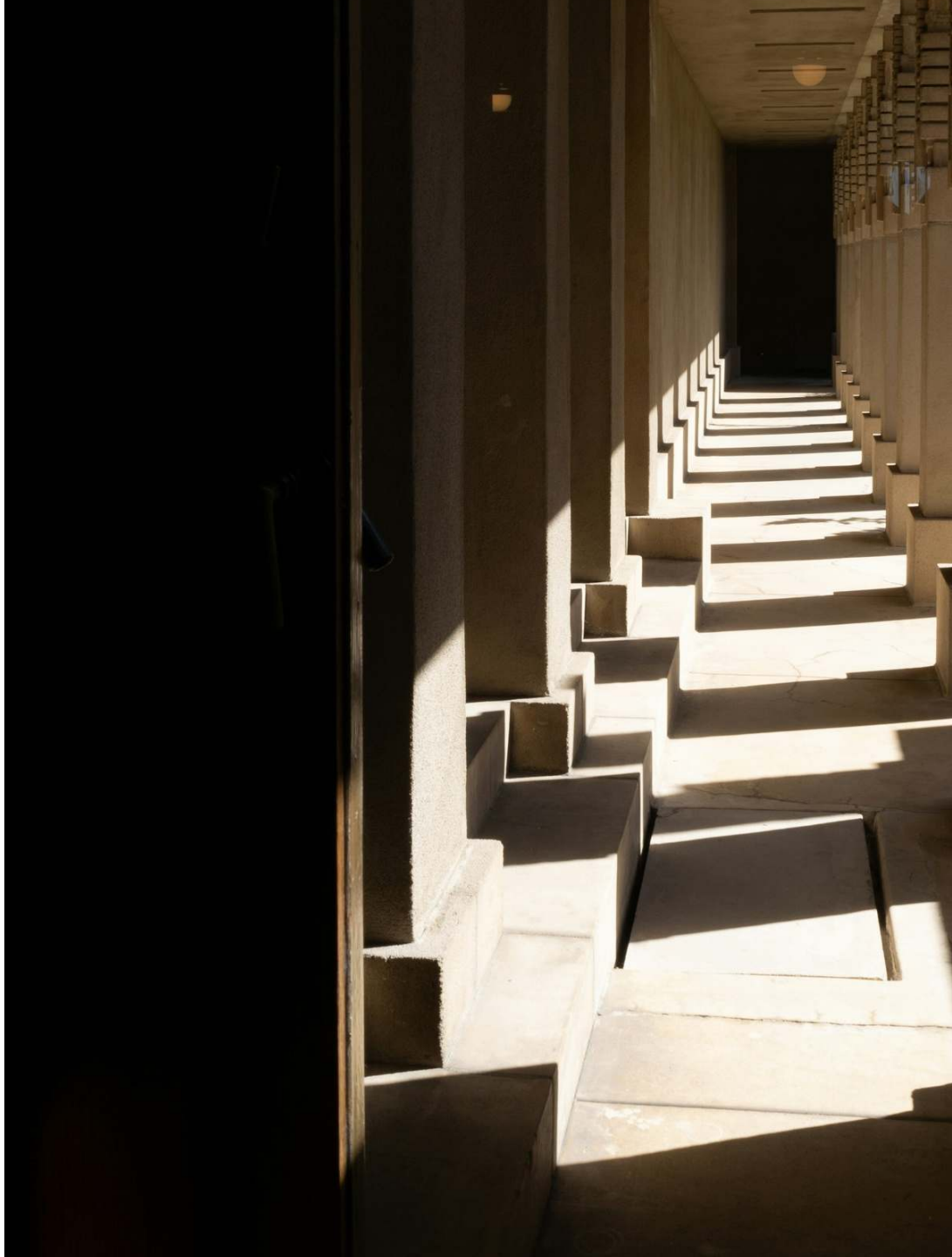
Rubric Self-Check

Narrative: rhythm → loss → fork.

Concept: **time** as spine; one proxy per scene.

Context: trade-offs + sovereignty sources.

Reflection: epistemic UI + user reflection + AI limits.



BDC Deliverables

10′ oral + 5′ Q&A · visuals / WebXR · minimal physical + QR

Creative Video

1–5′ duration

Slides

Available on Drive

Optional Tag

@biodesigned

Repository Pointers

Canonical Plan

BDC_2026_Extinction_Archive_Planning_Document.md

Executive Project Plan

PROJECT_PLAN.md

Reflection Template

templates/reflection-log-webxr.html

GitHub Preview

slides/SUMMIT_DECK_GITHUB_PREVIEW.md
(this file is auto-generated)

Thank You

Extinction Archive: Umwelt Hypothesis Dossiers